

The Future Unemployed

THE NEXT WAVE IN TECHNOLOGY

February 6, 2026
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Figure 1: Team Logo

Project Overview

Description

A client from a group of wealthy investors has approached us to create a system that simulates Panther Cloud Airlines' fleet operations over a two-week period. This platform will track and display real-time flight paths for each aircraft, showing their current positions and routes as they operate throughout the day. The system will calculate profitability for individual flights and routes by factoring in ticket sales, fuel costs, airport fees, and maintenance expenses. It includes a passenger load forecasting feature that predicts how full each flight will be based on the population of the metropolitan area. Additionally, the platform has an integrated booking system where users can search for flights, view available seats, and make reservations. All these components work together in real-time, when someone books a flight, the passenger count updates automatically, which then affects the profitability calculations, giving investors a complete view of the airline's operational performance and financial health.

Team Positions

- Reid Holloway - Scrum Master/Project Leader
- Graham McCabe - Configuration/Quality Assurance Manager
- Gracie Mitchell - Documentation Specialist

Work Breakdown

Reid Holloway

- Goal Hierarchy
- High-Level Cost Estimates
- Appendicies

Graham McCabe

- Implemented Database
- Operational Scenarios
- High-Level System Model
- User Types

Gracie Mitchell

- Functional and Non-Functional Requirements
- Created Database for Graham to Implement
- Worked on Designing Presentation
- Glossary

Project Goals

Financial Analysis

- Identify which routes and aircraft are most profitable to guide investment decisions
- Determine optimal pricing strategies across different fare classes and routes
- Project overall revenue and profit margins for the two-week operational period

Operational Efficiency Goals

- Optimize aircraft utilization rates to maximize flight hours and minimize ground time
- Identify underperforming routes that may need schedule adjustments or discontinuation
- Analyze fuel efficiency across different aircraft types and routes

System Demonstration Goals

- Showcase the technical capabilities of real-time simulation and integrated booking systems

Project Scope

Part 1

- Create a comprehensive timetable for an airline fleet operating across:
    - Top 30 US airports (FAA authorized)
    - 1 international route: New York ↔ Paris (daily)
    - Fleet:
        - 15 Boeing 737-600s
        - 15 Boeing 737-800s
        - 12 Airbus A220-100s
        - 13 Airbus A220-300s

Part 2

- Simulating 14-days of operation with different variables
    - Weather delays, icing delays, mechanical failures
    - Flight cancellations, gate delays
    - Jet stream effects
- Track all passenger movements, aircraft positions, gate usage
    - Capabilities:
        - Flight scheduling with realistic constraints
            - Gates
            - Maintenance
        - Route optimization based on
            - Distance
            - Range
            - Hub
            - Demand
        - Passenger journey
            - Connections
            - Profit
        - Aircraft data and usual performance
        - Finances
            - Costs
            - Revenue
            - Profit
        - Conversions to and from Paris
    - **NOT INCLUDED**
        - Crew Scheduling
        - Baggage Tracking
        - Multi-year simulation
        - Interaction with other airlines
        - Regulated airspace
    - Criteria for success

- All 56 aircraft scheduled properly
- All airports have ways from one to another
- All days simulate with no issues
- Profit is accurately output

Customer

Panther Cloud Air is a small group of wealthy investors who have decided to start their own airline. They are currently exploring the feasibility of their business plan and want to develop a simulation to see how robust it is, and to learn more about the logistics involved in running an airline. Their goal is to make a profit, and to provide excellent service at a reasonable price.

User Roles

Role	Permissions
Airport Employee	Log in, view scheduled flights, book flights for customers
Airport Administrator	All employee permissions plus access to route and plane profitability reports

Table 1: User Types

Functional and Non-Functional Requirements

Functional Requirements

- The system shall simulate a 14 day airline operation.
- The system shall track aircraft positions, flight paths, gate usage, and scheduled vs actual times, plus passenger movement across the system.
- The system shall support passenger journeys, including connections and ensure passengers reach the correct destination.
- The system shall allow employees to search for flights using origin, destination, and travel date.
- The system shall display scheduled flights including aircraft type, available seats, and departure time.

- The system shall allow employees to create bookings for passengers on available flights.
- The system shall automatically update passenger counts and seat availability when a booking is confirmed.
- The system shall calculate ticket prices using pricing rules and the configured profit margin.
- The system shall allow administrators to view profitability reports for routes and aircraft.

Non-Functional Requirements

- The system shall simulate the real-time calculations and must run fast enough to stay responsive even with updates like bookings and disruptions, within processing constraints.
- The system shall not crash during long runs within the 14 days, and should recover cleanly from errors to avoid data loss.
- The system shall have role-based access control and prevent employees from accessing admin-only profitability reports, and protect login credentials.

Operation Scenarios

Scenario 1: User logging in

1. User enters username and password
2. System queries Users WHERE username matches
3. System checks the password matches the stored hash
4. System reads the role field
5. If role = 'employee' → redirect to booking screen
6. If role = 'administrator' → redirect to admin dashboard
7. If credentials are wrong → show error message

Tables touched: Users

Scenario 2: Employee booking a flight

1. Employee enters origin, destination, date
2. System queries Flights WHERE status = 'scheduled' and matching route
3. Employee selects flight → system checks Aircraft_Models seat config minus count of confirmed Bookings

4. Employee selects seat class → system reads Pricing_rules and System_config to calculate price
5. Employee enters passenger info → system checks Passengers by email, creates new record if needed
6. Employee confirms → system inserts into Bookings

Tables touched: Flights, FlightDistances, Airports, Aircraft, Aircraft_Models, Pricing_rules, System_config, Passengers, Bookings

Scenario 3: Admin viewing profitability report

1. Admin logs in and lands on the admin dashboard
2. Admin navigates to the profitability report
3. System queries the Route_profitability view which automatically joins and aggregates data from Flights, FlightDistances, Airports, Ticket_prices, Flight_fuel_summary, and Bookings
4. System also queries Aircraft_profitability view which joins Aircraft, Aircraft_Models, Flights, Ticket_prices, and Flight_fuel_summary
5. UI displays results in a table showing revenue, fuel cost, and net profit per route and per aircraft

Tables touched: Route_profitability view, Aircraft_profitability view

Scenario 4: Admin setting profit margin

1. Admin navigates to admin dashboard
2. UI displays current profit margin read from System_config
3. Admin enters a new value (e.g. 15%)
4. System runs:

```
UPDATE System_config
SET config_value = 15.00
WHERE config_key = 'profit_margin'
```

5. UI confirms the update was saved
6. All future ticket price calculations automatically use the new margin

Tables touched: System_config

Goal Hierarchy

Level 1 - Overall Goal

- Create a working airline scheduling and 14-day simulation system for Panther Cloud Air that correctly models fleet operations, finances, and disruption scenarios

Level 2 - System Goals

1. Generate a complete, valid timetable for 56 aircraft across 31 airports including the Paris international route, stored in MariaDB and searchable by passengers
    - Weeks 1-7
        - Complete
2. Simulate 14 days of airline operations with all disruption scenarios, tracking every flight, aircraft, and gate
    - Week 11
        - Started
3. Produce end-of-simulation financial reporting: passengers transported, operating costs, revenue, and profit/loss
    - Week 12
        - Started
4. Deliver complete technical documentation including all appendices, README, installation guide, user manual, and code comments
    - Week 13
        - Not Started
5. Present and demonstrate the system live in a 20-minute team presentation with every member participating
    - Week 14
        - Not Started

Level 3 - Functional Goals

1.
    a. Airport & fleet data: GPS, metro populations, gate allocations, taxi times, 56 aircraft with tail numbers loaded into database
    b. Flight physics engine: Great Circle distance, bearing, wind adjustments (±4.5%), ascent/descent profiles, all validated with unit tests
    c. Scheduling algorithm: hub routing (ATL/DFW/DEN/ORD/JFK), gate allocation, 40/50 min turnarounds, operating hours 5am–1am enforced

d. Maintenance scheduling: every 200 flight hours at hub airports only, max 3 at the same time, aircraft end each day at same airport they start the next
e. Paris route: daily JFK↔CDG A350 service, Euro fuel/fee calculations, connecting passengers routed through JFK minimizing overnight layovers
f. Passenger demand model, search interface (A→B with connections), printed English timetable, ticket pricing at 30% load + configurable profit margin
    - Complete

2.

a. Simulation engine: day-by-day loop, aircraft state tracking, delay management, 30-min connection validation, all data logged to database
b. Disruption scenarios: weather delays (Day 3), icing above 40°N (Day 5), jet stream ±12% (Day 7), gate delays (Day 9)
c. Critical scenarios: aircraft failure at hub (Day 11), 8% flight cancellations west of 103°W with passenger rebooking (Day 13)
d. Post-simulation query interface: tail number tracking, on-time %, aircraft utilization, route and aircraft profitability views
    - Started

3.

a. Financial report: total passengers, fuel + airport fees + prorated lease costs, ticket revenue, profit/loss — broken down by route and aircraft
b. Admin dashboard with Route_profitability and Aircraft_profitability database views, role-based access control (employee vs administrator)
    - Started

4.

a. Airport data, aircraft specs, DB schema, hub rationale, tech stack, simulation schedule, formulas, currency conversion — drafted
    - Complete
b. README, installation guide, Makefile targets, user manual, code documentation, journal compliance — pending Part 2 completion
    - Not Started

5.

a. 20-minute presentation with live demo, all team members presenting, backup demo prepared, Q&A readiness
    - Not Started

Non-Functional Goals

- Performance
    - Simulation responsive throughout 14 days; booking updates propagate immediately to passenger counts and profitability
        - Started
- Security
    - Role-based access control, hashed passwords, GitHub repository remains private throughout (exposure = disqualification)
        - Complete
- Portability
    - Runs on client Dell Pro 14 (Windows 11) via Docker + docker-compose only; Makefile specifies Python 3.10+
        - Started
- Accuracy
    - Flight physics, wind adjustments, EUR/USD conversions all validated with unit tests and worked examples
        - Complete
- Reproducibility
    - All simulation random selections (delays, failures, cancellations) use a seeded RNG for consistent grading results
        - Not Started

System Overview

Panther Cloud Air is structured across four layers. A user interacts with HTML/CSS screens hosted on GitHub Pages. Each action triggers application logic that builds SQL queries, which execute against the MariaDB database running inside a Docker container.

Layer	Name	Technology	Responsibility
1.	User Interface	HTML/CSS	Displays screens and captures user inputs
2.	Application Logic	Query Builder	Validates inputs, builds queries, and applies pricing & fuel formulas

3.	SQL Operations	MariaDB	Executes SELECTs, INSERTs, JOINs against the database
4.	Database	MariaDB Docker	Persists all data; enforces foreign key constraints

Table 2: Layers

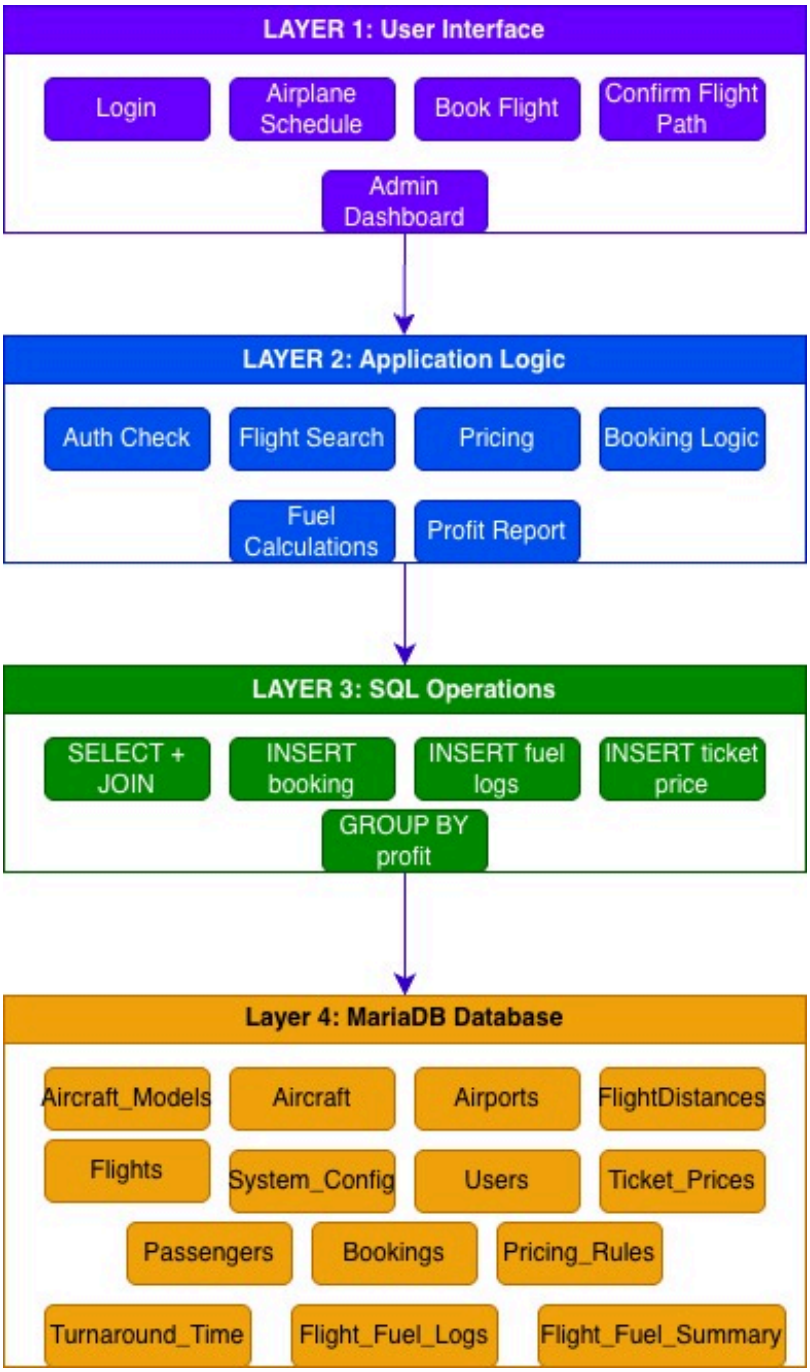


Figure 2: System Model

High-Level Cost Estimates

1. Fixed Monthly Costs (Fleet Leasing)
    - 15 × Boeing 737-600 @ $245,000/month = **$3,675,000**
    - 15 × Boeing 737-800 @ $270,000/month = **$4,050,000**
    - 12 × Airbus A220-100 @ $192,000/month = **$2,304,000**
    - 13 × Airbus A220-300 @ $228,000/month = **$2,964,000**

    Total monthly fleet cost: ~**$12,993,000**

2. Operation Costs per Flight

Aircraft	**Seat Capacity**	**Max Range**	**Fuel Cost / Hour @ $6.19/gal**
Boeing 737-600	132 passengers	3,050 nm	~$3,714 per hour
Boeing 737-800	162 passenger	2,940 nm	~$5,261 per hour
Airbus A220-100	133 passengers	3,400 nm	~$2,786 per hour
Airbus A220-300	160 passengers	3,400 nm	~$3,095 per hour
Airbus A350	325 passengers	8,500 nm	~$8,666 per hour

Table 3: Operation Costs per Flight

3. Extra Fuel Costs
    - US domestic fuel price: $6.19/gallon (locked in at fixed 1-year rate)
    - Paris fuel price: €1.97/liter (will be converted to USD monthly using rate from last day of month)
4. Landing/Takeoff Fees
    - US airport fees: $2,000 per takeoff/landing
    - Paris takeoff/landing fees: €2,100 per operation
        i. These are billed to the airline at the end of each month

Revenue Model

1. Passenger Demand Formula
    - 0.5% of metro population travels by air on any given day
    - Panther Cloud Air starts with 2% market share, growing to 5% by year end
    - Demand to any specific destination airport = (destination metro pop ÷ total reachable metro pop) × total daily travelers × market share
2. Ticket Pricing
    - Prices are calculated assuming only 30% of seats are filled
        - this is the break-even floor
    - All costs (fuel, fees, leasing prorated per flight) are divided across that 30% seat assumption
        - Profit margin is added on top
    - The goal is to reach 75% seat occupancy in practice
        - This would generate significant profit above the break-even price
3. Paris International Route
    - Fuel billed in euros at €1.97/liters
        - This requires liter-to-gallon conversion AND currency conversion
    - Landing/takeoff fees billed in euros at €2,100 each
    - The aircraft used for the Paris route will need to be capable of transatlantic flight
        - This will be the 737-800, A220-300, or (most likely) the A350
    - Passengers connecting from other US cities to Paris must route through JFK in New York

Glossary

1. Aircraft: A plane used by Panther Cloud Airlines to transport passengers between airports.
2. Administrator: A user with elevated permissions who can view reports and modify system settings.
3. Airport Employee: A system user who can search for flights and book tickets for passengers.
4. Booking: A reservation created for a passenger on a specific flight.
5. Database: A structured system used to store and manage data such as flights, passengers, and bookings.
6. Flight: A scheduled trip of an aircraft from one airport to another.

7. Passenger: A person traveling on a booked flight.
8. Profit Margin: The percentage added to operational costs to determine ticket prices.
9. Route: The path between two airports that a flight follows.
10. Simulation: A model that imitates airline operations over a set period of time.
11. Ticket Price: The cost paid by a passenger for a seat on a flight.
12. User: Any individual who interacts with the system, including employees and administrators.

Appendices

Appendix A - Top 30 US Airports

City/State	IATA Code	Airport Name	Metro Population	Hub?	Gates Available
Atlanta, Georgia	ATL	Hartsfield-Jackson Atlanta International	6,411,000	Yes	11
Dallas, Texas	DFW	Dallas/Fort Worth International	7,637,000	Yes	11
Denver, Colorado	DEN	Denver International	2,963,000	Yes	11
Los Angeles, California	LAX	Los Angeles International	13,200,000	No	5
Chicago, Illinois	ORD	O'Hare International	13,200,000	Yes	11
New York, New York	JFK	John F. Kennedy International	19,500,000	Yes	11
Orlando, Florida	MCO	Orlando International	2,673,000	No	3
Las Vegas, Nevada	LAS	Harry Reid International	2,265,000	No	2
Charlotte, North	CLT	Charlotte Douglas International	2,797,000	No	3

Carolina					
MIami, Florida	MIA	Miami International	6,183,000	No	5
Seattle, Washington	SEA	Seattle-Tacoma International	4,012,000	No	4
New York, New York	EWR	Newark Liberty International	19,500,000	No	5
Phoenix, Arizona	PHX	Phoenix Sky Harbor International	4,948,000	No	5
Houston, Texas	IAH	George Bush Intercontinental	7,122,000	No	5
Boston, Massachusetts	BOS	Logan International	4,919,000	No	5
Fort Lauderdale, Florida	FLL	Fort Lauderdale-Hollywood International	6,183,000	No	5
St. Paul, Minnesota	MSP	Minneapolis-St. Paul International	3,690,000	No	4
New York, New York	LGA	LaGuardia	19,500,000	No	5
Detroit, Michigan	DTW	Detroit Metropolitan Wayne County	4,365,000	No	4
Philadelphia, Pennsylvania	PHL	Philadelphia International	6,245,000	No	5
San Francisco, California	SFO	San Francisco International	4,623,000	No	5
Salt Lake City, Utah	SLC	Salt Lake City International	1,257,000	No	1
Washington DC	DCA	Ronald Reagan Washington National	6,385,000	No	5
Washington	BWI	Baltimore/Washingt	6,385,000	No	5

DC		on International			
Washington DC	IAD	Washington Dulles International	6,385,000	No	5
Tampa, Florida	TPA	Tampa International	3,342,000	No	3
San Diego, California	SAN	San Diego International	3,269,000	No	3
Portland, Oregon	PDX	Portland International	2,512,000	No	3
Honolulu, Hawaii	HNL	Honolulu International	974,000	No	1
Austin, Texas	IAH	Austin-Bergstrom International	2,352,000	No	2

Table 3: All 30 US Airports

Appendix B - Hub Selection Rationale

Hubs: ATL, DFW, DEN, ORD, and JFK.

1. Hub Criteria
    - Geographic centrality
        - Hubs should minimize average connection distance for passengers travelling between any two US airports
    - Metro population
        - Larger populations create more local demand, making the hub worthwhile to maintain at full 11-gate capacity
    - Route coverage
        - Each hub should serve a distinct geographic region to avoid overlap
    - Maintenance capacity
        - Hubs host up to 3 aircraft simultaneously for maintenance; selecting well-distributed hubs reduces long flight distances
    - International gateway requirement
        - The Paris route mandates a dedicated East Coast hub capable of hosting the A350 and minimizing layovers for connecting passengers

2. Selected Hubs
    - Atlanta
        - This airport's Southeast location makes it the go-to connecting point for all East Coast traffic not starting in New York. It has a low average distance to Southeast and Mid-Atlantic airports.
    - Dallas-Fort Worth
        - This airport's central location in the southern US covers Texas, the Gulf Coast, and Southwest routes. It reduces number of stops required for passengers travelling between the Southeast and West Coast.
    - Denver
        - It is in the geographic center of the continental US. It is the critical East-West connector, ensuring that no cross-country passenger needs more than one connection. It covers Mountain region airports and provides a consistent port for Western routes.
    - Chicago O'Hare
        - ORD covers the entire Midwest and provides a North-South connections between Canadian border cities and the East coast. It also prevents Denver from being overloaded with Midwest connections.
    - New York JFK
        - JFK is mandatory for the daily New York-Paris A350 service. All passengers travelling to Paris from other US cities must connect through JFK and being a hub ensures gate availability for both domestic flights and the international.